# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

# CHEMISTRY

Past Paper 2023 • Paper Code: 9231

Retyped faithfully from the original board paper — same questions, no changes.

## SECTION-A

**Time Allowed:** 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** 9231

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. When gas diffuse, it tend to equalize the:

(A) Partial pressure
(B) Volume
(C) Area
(D) Temperature

Q2. What is molarity of solution which contain 40 g sodium chloride dissolved in 500 $cm^3$?

(A) 2.36 M
(B) 1.36 M
(C) 0.36 M
(D) 0.16 M

Q3. A heterogeneous solution in which the size of particles are 1-$10^3$ nm and cannot be separated on standing is called:

(A) Colloidal
(B) Suspension
(C) Solid Solution
(D) Aqueous Solution

Q4. Which of the following is not a reducing agent?

(A) $NaNO_2$
(B) $NaNO_3$
(C) HI
(D) $SnCl_2$

Q5. What is the oxidation number of manganese in $MnO_4^-$?

(A) +4
(B) −4
(C) +7
(D) −7

Q6. In periodic table, metallic character of elements starts decreasing from:

(A) Left to Right in periods.
(B) Right to Left in periods.
(C) Top to Bottom in groups.
(D) Bottom to Top in groups.

Q7. Chemical name of bleaching powder is:

(A) Calcium Carbonate
(B) Calcium Oxide
(C) Calcium Hydroxide
(D) Calcium Hypochlorite

Q8. Free radicals are formed by?

(A) Electrolytic fission
(B) Hetrolytic fission
(C) Homolytic fission
(D) Induced fission

Q9. The size of gold foil which was used in Rutherford's experiment:

(A) 0.04 cm
(B) 0.004 cm
(C) 0.0004 cm
(D) 0.00004 cm

Q10. In periodic table group number indicates the number of:

(A) Electrons in valence shell
(B) Protons in nucleus
(C) Neutrons in nucleus
(D) Number of shells

Q11. Sodium has low ionization energy so it will form?

(A) Cation
(B) Radical
(C) Anion
(D) Negative Ion

Q12. The type of Covalent present in Carbon dioxide is:

(A) Single Covalent Bond
(B) Double Covalent Bond
(C) Triple Covalent Bond
(D) Co-ordinate Covalent Bond

## SECTION-B

**Time Allowed:** 2 Hours 40 Minutes | **Maximum Marks:** 32 | **Paper Code:** 9231

Q1. Attempt any eight of the following short questions. Each question carries 4 marks

(i) How many moles are in 50g of Carbon and 40g of Oxygen?

(ii) Write at least two differences between proton and neutron?

(iii) Sketch the shape of periodic table and mark off those areas where we find? (i) s-block (ii) p-block (iii) d-block (iv) f-block

(iv) (i) Why do atom form bonds? (ii) In how many ways atoms form bonds?

(v) With the help of dot cross model show the formation of: (i) $N_2$ molecule (ii) $NH_3$ molecule

(vi) A balloon filled with air in a room with temperature at 22.0°C has a volume of 50.2 ml. The balloon is taken outside where temperature is 36.6°C. What will be volume of balloon after.

(vii) Identify Solute and Solvent in Sterling Silver and Brass.

(viii) What volume of water is added in 90$cm^3$ of 2.5 Mole HCl solution to make 0.5 molar solution?

(ix) Write down the half cell reactions in Nelson Cell.

(x) Define electroplating and give three purposes of electroplating.

(xi) (i) Why is platinum used for making jewelry? (ii) Why it is considered as expensive metal?

## SECTION-C

**Maximum Marks:** 21

Note: Attempt any three of the following questions. Each question carries 7 marks

Q2.

(i) Calculate formula mass of $Fe(NO_3)_3$ in a.m.u and express it in grams also?

(ii) Draw the structure of: $^{12}_{6}C$, $^{14}_{6}C$, $^{1}_{1}H$ and $^{2}_{1}H$.

Q3.

(i) Why in same group, elements shows resemblance in chemical properties? Justify answer with example.

(ii) What is metallic bonding. Explain the theory which is related to metallic bond.

Q4.

(i) Find volume of a 3 $dm^3$ gas if pressure is increased from 400 mm of Hg to 800 mm of Hg.

(ii) Define saturated solution elaborate your answer with suitable example?

Q5.

(i) Write reaction of sodium with hydrogen and oxygen?

(ii) Write half cell reactions in Daniel cell also write overall reaction.